

MySQL Assignment 2 – Querying Data

Database & Table Creation

Created the database and tables (departments, location, and employees) based on the Assignment 1 schema.

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane lists the databases: emp, employee, schooldb, sys, and testdb. The 'employee' database is selected. In the center, the SQL editor shows the following commands:

```
1 • DROP DATABASE IF EXISTS employee;
2 • CREATE DATABASE employee;
3 • USE employee;

4
5 • CREATE TABLE departments (
6     department_id INT PRIMARY KEY,
7     department_name VARCHAR(100) NOT NULL UNIQUE
8 );
9
10 • CREATE TABLE location (
11     location_id INT PRIMARY KEY AUTO_INCREMENT,
12     location VARCHAR(30) NOT NULL UNIQUE
13 );
```

Populated the tables by executing the provided DML insert script

The screenshot shows the MySQL Workbench interface with the 'employee' database selected. The SQL editor displays the following commands:

```
51 (10, 'Customer Support'), (11, 'Business Development'), (12, 'IT');
52
53 • select * from employees;
54
55 • INSERT INTO location (location) VALUES
```

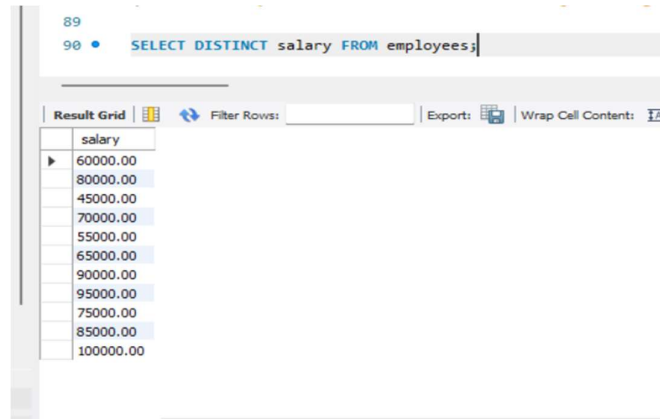
Below the SQL editor, the 'Result Grid' shows the data in the 'employees' table:

employee_id	employee_name	gender	hire_date	designation
5001	Vihaan Singh	M	2015-01-20	Data Analyst
5002	Reyansh Singh	M	2015-03-10	Network Engineer
5003	Aaradhya Iyer	F	2015-05-20	Customer Support Executive
5004	Kiara Malhotra	F	2015-07-05	Business Development Executive
5005	Anvi Chaudhary	F	2015-09-11	UI Developer
5006	Dhruv Shetty	M	2015-11-20	Marketing Manager
5007	Anushka Singh	F	2016-01-15	Graphic Designer
5008	Diya Jha	F	2016-03-05	Sales Executive
5009	Kaav Desai	M	2016-05-20	Systems Administrator
5010	Atharv Yadav	M	2016-07-10	Marketing Analyst
5011	Saanvi Patel	F	2016-09-20	Operations Manager
5012	Myra Verma	F	2016-11-05	Customer Success Manager
5013	Arnav Rao	M	2017-01-20	

Executed all the SQL queries required

1. Distinct Values:

🕒 a query to retrieve distinct salaries from the Employees table

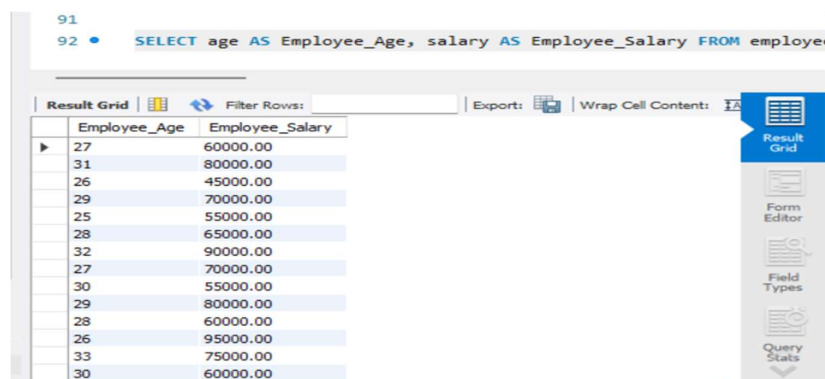


The screenshot shows a SQL query editor with the query: `SELECT DISTINCT salary FROM employees;`. Below the query, the 'Result Grid' displays a list of distinct salary values.

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

2. Alias (AS):

🕒 Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

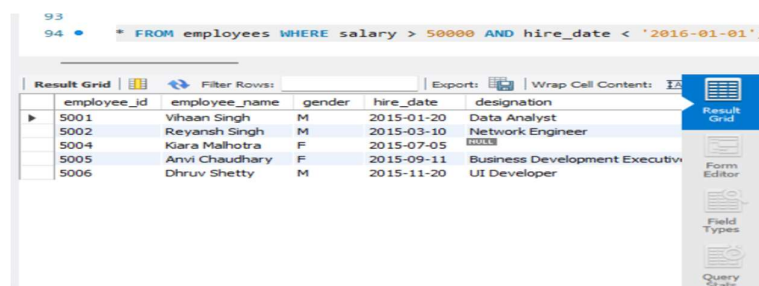


The screenshot shows a SQL query editor with the query: `SELECT age AS Employee_Age, salary AS Employee_Salary FROM employees;`. Below the query, the 'Result Grid' displays the results with the specified aliases.

Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00

3. Where Clause & Operators:

🕒 Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.



The screenshot shows a SQL query editor with the query: `* FROM employees WHERE salary > 50000 AND hire_date < '2016-01-01';`. Below the query, the 'Result Grid' displays the results for employees meeting the criteria.

employee_id	employee_name	gender	hire_date	designation
5001	Vihaan Singh	M	2015-01-20	Data Analyst
5002	Reyanish Singh	M	2015-03-10	Network Engineer
5004	Kiara Malhotra	F	2015-07-05	Business Development Executive
5005	Anvi Chaudhary	F	2015-09-11	Business Development Executive
5006	Dhruv Shetty	M	2015-11-20	UI Developer

☉ Find the employee whose designation is missing and fill it with "Data Scientist".

```
95
96 • UPDATE employees SET designation = 'Data Scientist' WHERE designation IS NULL;
```

employee_id	employee_name	gender	hire_date	designation	department_id	location_id	salary	email	age
5001	Vishaan Singh	M	2015-01-20	Data Analyst	3	4	60000.00	NULL	27
5002	Reyansh Singh	M	2015-03-10	Network Engineer	12	1	80000.00	NULL	31
5004	Kiara Malhotra	F	2015-07-05	NULL	8	3	70000.00	NULL	29
5005	Anvi Chaudhary	F	2015-09-11	Business Development Executive	11	1	55000.00	NULL	25
5006	Dhruv Shetty	M	2015-11-20	UI Developer	8	2	65000.00	NULL	28

1. ORDER BY:

☉ Find employees sorted by department ID in ascending order and salary in descending order.

2. LIMIT:

☉ Display the first 5 employees hired in the year 2018

```
102 • BETWEEN '2018-01-01' AND '2018-12-31' ORDER BY hire_date ASC LIMIT 5;
```

employee_id	employee_name	gender	hire_date	designation	department_id	location_id	salary	email	age
5019	Ishika Patel	F	2018-01-15	Talent Acquisition Specialist	4	4	55000.00	NULL	29
5020	Aarav Nair	M	2018-03-05	Software Engineer	1	1	90000.00	NULL	32
5021	Advik Kapoor	M	2018-05-20	Finance Analyst	7	3	85000.00	NULL	26
5022	Aadhya Iyengar	F	2018-07-10	HR Specialist	4	4	60000.00	NULL	28
5023	Anika Paul	F	2018-09-20	Public Relations Specialist	2	2	70000.00	NULL	30

3. Aggregate Functions:

☉ Calculate the sum of all salaries in the Finance department.

```
103
104 • SELECT SUM(salary) FROM employees WHERE department_id = 7;
105
```

SUM(salary)
170000.00

④ Find the minimum age among all employees

```
106 • SELECT MIN(age) FROM employees;
107
```

Result Grid | Filter Rows: | Export: | Wrap

MIN(age)
25

4. GROUP BY:

④ List the maximum salary for each location.

```
108 • SELECT location_id, MAX(salary) FROM employees GROUP BY 1
109
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	MAX(salary)
1	95000.00
2	95000.00
3	90000.00
4	100000.00

max

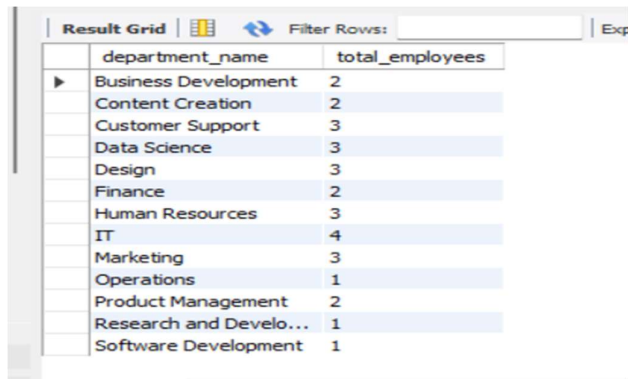
④ Calculate the average salary for each designation containing the word 'Analyst'.

Result Grid | Filter Rows: | Export: | V

designation	AVG(salary)
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

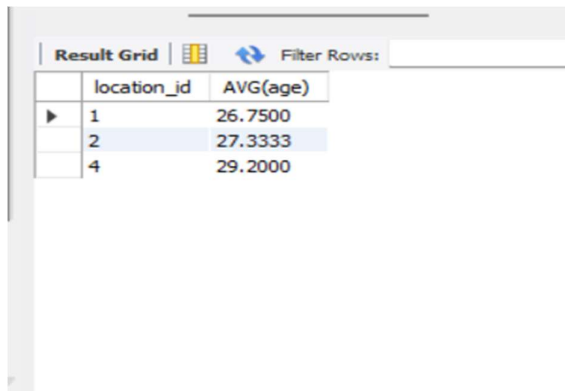
🕒 Find departments with less than 3 employees.



The screenshot shows a 'Result Grid' with two columns: 'department_name' and 'total_employees'. The data is as follows:

department_name	total_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Development	1
Software Development	1

🕒 Find locations with female employees whose average age is below 30.



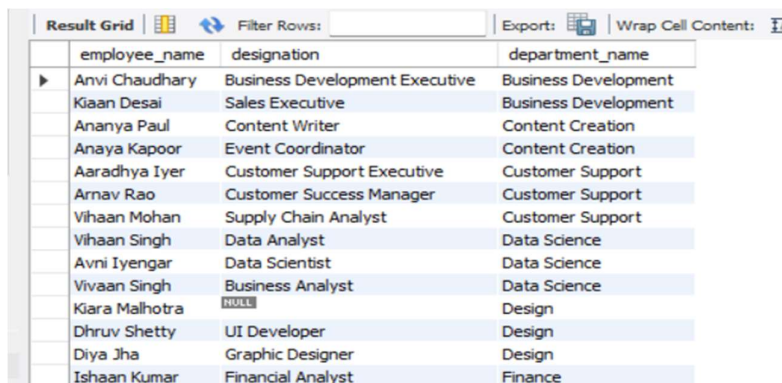
The screenshot shows a 'Result Grid' with two columns: 'location_id' and 'AVG(age)'. The data is as follows:

location_id	AVG(age)
1	26.7500
2	27.3333
4	29.2000

Joins:

1. Inner Join:

🕒 List employee names, their designations, and department names where employees are assigned to a department

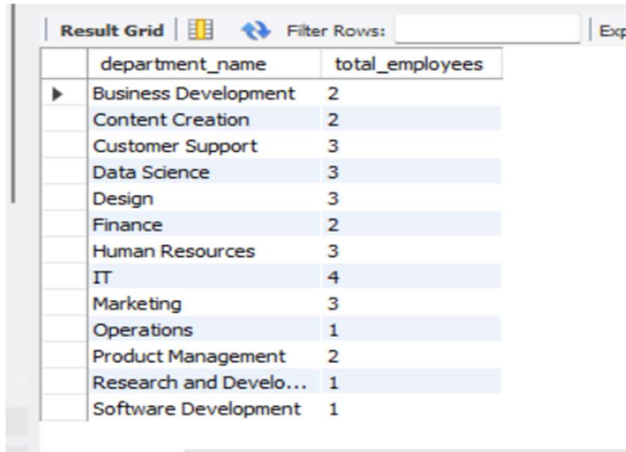


The screenshot shows a 'Result Grid' with three columns: 'employee_name', 'designation', and 'department_name'. The data is as follows:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Kiara Malhotra	UI Developer	Design
Dhruv Shetty	UI Developer	Design
Diya Jha	Graphic Designer	Design
Ishaan Kumar	Financial Analyst	Finance

2. Left Join:

🕒 List all departments along with the total number of employees in each department, including departments with no employees.

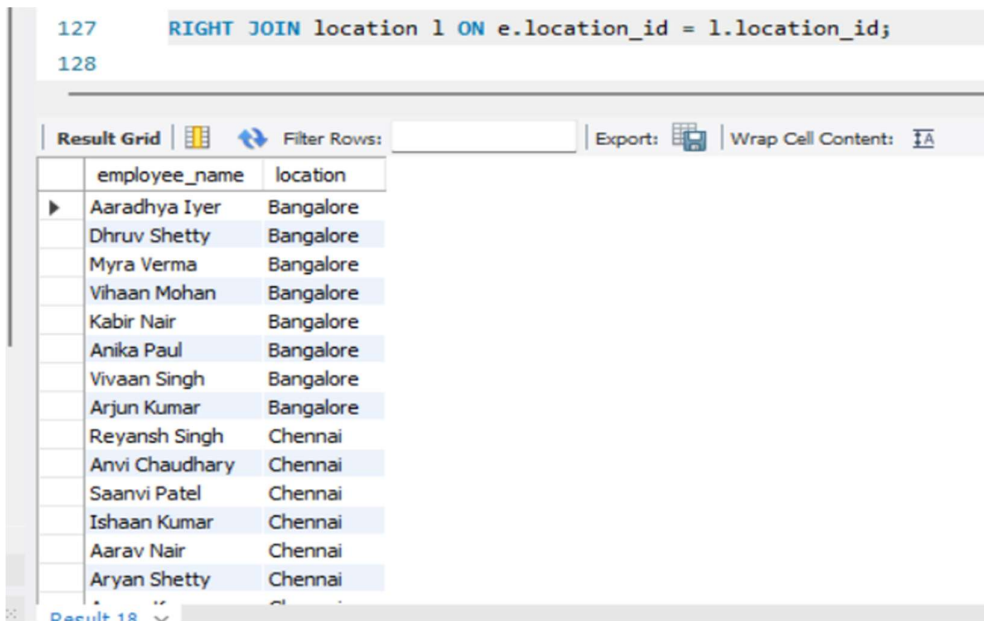


The screenshot shows a SQL query result grid with two columns: 'department_name' and 'total_employees'. The grid lists 15 departments and their corresponding employee counts. The departments are: Business Development (2), Content Creation (2), Customer Support (3), Data Science (3), Design (3), Finance (2), Human Resources (3), IT (4), Marketing (3), Operations (1), Product Management (2), Research and Development (1), and Software Development (1).

department_name	total_employees
Business Development	2
Content Creation	2
Customer Support	3
Data Science	3
Design	3
Finance	2
Human Resources	3
IT	4
Marketing	3
Operations	1
Product Management	2
Research and Development	1
Software Development	1

3. Right Join:

🕒 Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.



The screenshot shows a SQL query result grid with two columns: 'employee_name' and 'location'. The grid lists 18 employees and their assigned locations. The employees are: Aaradhya Iyer, Dhruv Shetty, Myra Verma, Vihaan Mohan, Kabir Nair, Anika Paul, Vivaan Singh, Arjun Kumar, Reyansh Singh, Anvi Chaudhary, Saanvi Patel, Ishaan Kumar, Aarav Nair, and Aryan Shetty. The locations are: Bangalore (10 employees) and Chennai (4 employees).

employee_name	location
Aaradhya Iyer	Bangalore
Dhruv Shetty	Bangalore
Myra Verma	Bangalore
Vihaan Mohan	Bangalore
Kabir Nair	Bangalore
Anika Paul	Bangalore
Vivaan Singh	Bangalore
Arjun Kumar	Bangalore
Reyansh Singh	Chennai
Anvi Chaudhary	Chennai
Saanvi Patel	Chennai
Ishaan Kumar	Chennai
Aarav Nair	Chennai
Aryan Shetty	Chennai

THANK YOU